

Evaluation Metrics for Classification

- Confusion Matrix

Why not just use Accuracy? Accuracy can be misleading, especially with imbalanced datasets. We need deeper insights into model performance.

Confusion Matrix

→ Binary Classification (2×2 Matrix)

		Predicted	
		0	1
Actual	0	TN True Negative	FP False Positive (Type 1 Error)
	1	FN False Negative (Type 2 Error)	TP True Positive

e.g., Plagiarism detection: Flagging original work as plagiarized.

e.g., Manufacturing: Missing a defective product.

For Multi-Class Classification: Extends to an ' $n \times n$ ' matrix.

Evaluation Metrics – Continued

1. Accuracy:

How often is the model correct overall?

$$\rightarrow \text{Accuracy} = \frac{\text{TN} + \text{TP}}{\text{TN} + \text{FP} + \text{FN} + \text{TP}}$$

		Predicted	
		0	1
Actual	0	TN	FP
	1	FN	TP

Limitation: Can be misleading for imbalanced datasets (e.g., 99% non-spam).

2. Precision: When model predicts positive, how often is it correct?

$$\rightarrow \text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

Intuition: Focuses on minimizing false alarms (e.g., not labeling non-spam as spam).

3. Recall (Sensitivity): How many actual positives did the model catch?

$$\rightarrow \text{Recall} = \frac{\text{TP}}{\text{FN} + \text{TP}}$$

Intuition: Focuses on minimizing missed positives (e.g., not missing a fraudulent transaction).

4. F1 Score: Harmonic mean of Precision & Recall.

$$\rightarrow \text{F1 Score} = \frac{2 * \text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}}$$

Why useful? Good for imbalanced data; balances the trade-off between Precision and Recall.
